
Predicted Impacts of a Change of NAP Food Stamps on Labor Market Outcomes in Puerto Rico

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Abstract

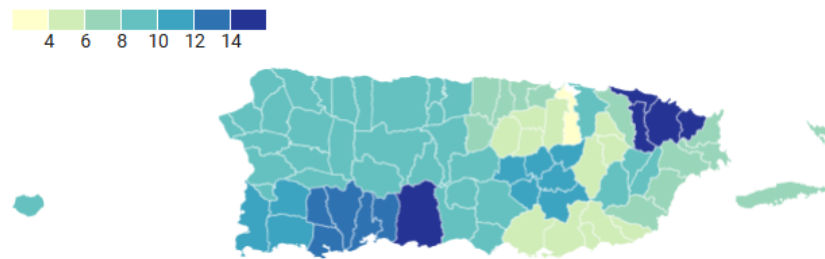
Puerto Rico is a United States territory and receives direct government funding from the United States. However, the welfare programs in place in Puerto Rico still remain different than those of the United States. In particular, the Puerto Rican equivalent of the U.S. Supplemental Nutrition Program (SNAP) is the Nutrition Assistance Program (NAP). SNAP is has more flexible income eligibility criteria and provides higher benefits on average as compared to NAP. Given this, there has been much debate about whether it would be beneficial for Puerto Rico to switch to SNAP. This paper discusses the impacts of a potential switch to SNAP on the an individual's decision of employment and their decision of whether to receive benefits given they are eligible for the program. The average elasticity of employment, average elasticity of weekly hours worked, and average elasticity of benefit take up in response to a change in benefit amount is found to be -0.02518 , -0.00043 , and 0.03128 respectively. Further, a transition from NAP to SNAP is predicted to increase Puerto Rico's overall unemployment rate by 4.1898%, decrease weekly hours worked by -0.3352% , and increase benefit take up rate by 1.8891%. This translates to 54,926 less individuals being employed, 154,426 less weekly hours worked by the population, and 15,941 more individuals receiving benefits under SNAP. Assuming a casual and linear relationship between benefit amounts and employment decisions, the \$1,337 increase in average yearly benefit amounts under SNAP, is likely to disincentivize employment.

Introduction

Puerto Rico is a territory of the United States and has historically experienced higher rates of unemployment and poverty than the mainland U.S.. In 2024, GDP per capita was \$39,343 for Puerto Rico and \$84,534 for the United States, representing a large divide in terms of economic well being (Federal Reserve Bank of St. Louis). Further, all 78 municipalities in Puerto Rico had higher poverty rates than Louisiana, the U.S. state with the highest

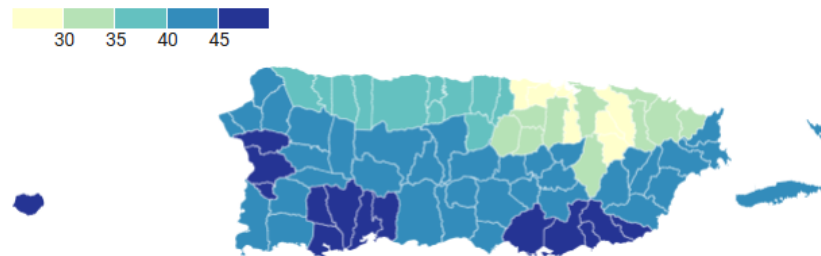
poverty rate of 18.7%, where the smallest poverty rate for a municipality was 25.9% (for Dorado and Toa Baja). This is coupled with an average unemployment rate of 5.6% in Puerto Rico in 2024, as compared to the 4.4% nationwide average for the United States (U.S. Bureau of Labor Statistics). The variation in 2024 unemployment rate and poverty rate across municipalities in Puerto Rico is displayed in Figures 1 and 2.

Fig 1. Variation in Unemployment Rate in 2024 Across Municipalities in Puerto Rico



Map: Tamara Gruslova • Source: Puerto Rico Community Survey 2024 • Created with Datawrapper

Fig 2. Variation in Poverty Rate in 2024 Across Municipalities in Puerto Rico



Map: Tamara Gruslova • Source: Puerto Rico Community Survey 2024 • Created with Datawrapper

Considering these financial hardships that Puerto Rico experiences, maintaining successful welfare programs is top of mind for providing a support system for families and allowing individuals to get back on their feet. One of these influential welfare programs is food stamps, which are crucial for providing access to healthy, nutritious food, especially for children. The current food stamps program in place in Puerto Rico is the Nutrition Assistance Program (NAP) or in Spanish, Programa de Asistencia Nutrición (PAN). NAP is a block grant

provided yearly by the United States government, where the amount allocated for NAP is fixed and irrespective of the number of needy recipients. However, given that Puerto Rico is a United States territory granting U.S. citizenship to all Puerto Ricans, there has been discussion about why Puerto Rico has a separate food stamps program from the U.S..

The food stamps program in the mainland United States is the Supplemental Nutrition Assistance Program (SNAP). Contrastingly to NAP, SNAP is a federal program that provides funding based on the amount of individuals in need. This means that SNAP is able to expand its program during periods of economic downturn or natural disasters to ensure that all eligible citizens receive aid. Further, this allows SNAP to have more generous benefit amounts and less restrictive eligibility criteria on average, as compared to NAP. Given this seemingly obvious advantage to SNAP, the question of whether a transition from NAP to SNAP would lower financial hardship in Puerto Rico arises.

There have been several policy briefs over the past five years that aim to understand this question and advocate for, at least, a critical examination of the effectivity of NAP in Puerto Rico. In 2010, the SNAP feasibility study conducted by Peterson et. al with the United States Department of Agriculture found that SNAP would increase the number of individuals eligible and actively participating in the program by 15.3%, increase the population coverage from 30% to 43%, and increase average monthly benefits by 9.6%. A July 2022 update of this study by Thorn et. al further touched on the implementation costs and necessary steps for the Puerto Rican government to take if SNAP were to be adopted. They find that full implementation of SNAP would take around 10 years to complete, incur \$341-426 million in implementation costs, have 861,000 participating households, and pay \$4.5 billion in benefits annually. They further discuss how the NAP system is too outdated to adapt into the SNAP program and thus, how a transition to SNAP would require implementation of entirely new systems (including SNAP Employment & Training program, Disaster SNAP, and SNAP Ed among others) in order to comply with SNAP program requirements. They conclude that transitioning to SNAP is feasible and would have meaningful impacts on eligibility and benefit amounts, but would need extensive and costly reorganization directed by the Puerto Rican government in order for successful implementation.

In a separate study, Marxuach (2022) finds that a transition to SNAP would have the following positive effects: the number of beneficiaries would increase from 9-12%, maximum benefit amounts would increase, and increased spending in grocery items would sustain a positive income multiplier effect on the local economy. They do note the downsides of high implementation and administration costs and some individuals losing access to benefits, however, they conclude with the fact that a transition to SNAP would be largely beneficial to Puerto Rico. Another paper by Carrión-Tavárez (2022) highlights that the less restrictive eligibility criteria under SNAP as helping to avoid the "poverty trap" caused by NAP, which is defined to occur when individuals' costs of working are too high given that they will lose their benefits if they exceed the eligible income level, and that this would likely increase labor force participation.

However, none of these studies quantitatively measure the impacts a transition to SNAP on labor behavior. Valdés Valderrama (2018) finds that an increase in the number of households receiving NAP benefits caused a significant negative impact on employment rate, though of a small magnitude. Further, Hoynes and Schanzenbach (2018) found that the introduction of food stamps in the United States also reduced the employment and hours worked among the population. Given this, it is interesting to measure whether a hypothetical transition to SNAP would align with this economic intuition of increased benefits disincentivize employment or whether program eligibility changes with improve labor participation if individual's can still receive benefits while they hold a job. This paper aims to measure the changes in eligibility, benefit amounts, and benefit takeup between NAP and SNAP, and the impacts of a transition to SNAP on labor market outcomes, including individual probability of employment and the number of hours worked weekly.

Data

All data for the employment and benefit take up analysis was taken from the 2023 Puerto Rico Community Survey Public Use Microdata Sample (PRCS PUMS). The PRCS PUMS is an extension of the U.S. Census Bureau's Puerto Rico Community Survey (PRCS), providing individual and household level observations. All 27,427 observations correspond to a single individual who belongs to a particular household which can be determined using the individual's household ID. For each of these individuals, information on their personal char-

acteristics, employment status, income, welfare receipt, and their individual survey weight was taken. The dataset contains 14,696 men and 12,731 women, of which 4,611 [442] men and 4,479 [425] women are employed [unemployed] respectively. In total, 9,090 individuals are employed, 867 are unemployed, and the rest are either not part of the labor force or under 16 years old. Figures 3 and 4 display the distribution of employed and unemployed status by age for men and women combined. We see that employment is largest between age 40-50 and is decreasing on both sides of this peak, while unemployment is largest amount young adults aged between 18-25 and is decreasing with increasing age.

Figure 3. Count of Employed Individuals by Age

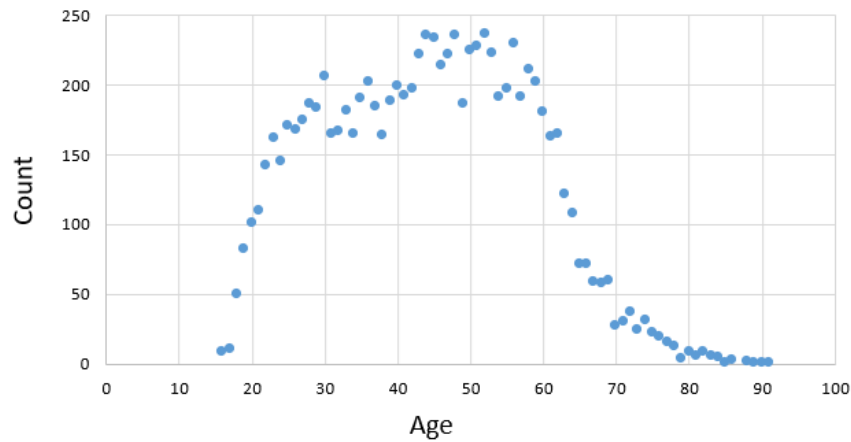
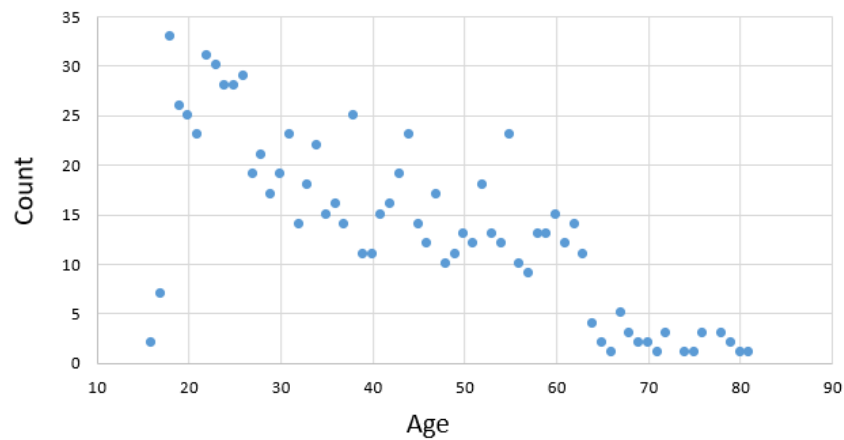
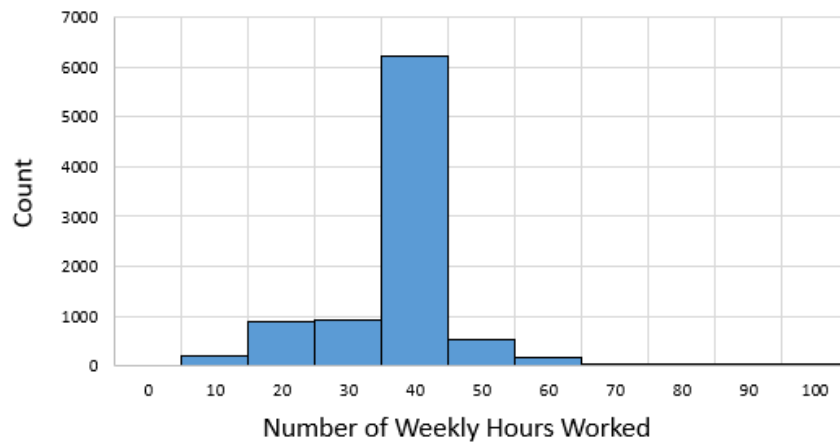


Figure 4. Count of Unemployed Individuals by Age



Further, the distribution of number of hours worked per week conditional on an individual being employed is displayed in Figure 5. Majority of employed individuals work full time, with there being 5,266 individuals working 40 hours a week, though there was a considerable amount of individuals that work part-time, with 1516 individuals working between 20 to 30 hours. It is important to consider the fact that most individuals work full time, given that further analysis is centered on the individual decision to become employed and in this case, work full time.

Figure 5. Histogram of Weekly Hours Worked



Additional data on the unemployment rate, food stamp receipt, and number of businesses authorized to accept NAP food stamps across municipalities was used. Data on the size of the labor force and unemployment counts for each municipality in 2023 was collected from the Local Area Unemployment Statistics program of the Puerto Rico Department of Labor and Human Resources. These counts were aggregated across municipalities to adjust the unemployment rate to the PUMA level. Data on the number of individuals receiving NAP food stamps in 2022 was collected from the 2022 Puerto Rico Community Survey Public Use Microdata Sample (PRCS PUMS). Data on businesses certified for accepting NAP food stamps was collected from the Puerto Rico Administration of the Socioeconomic Development of the Family (ADSEF).

As a test of the consistency of the study's findings, the employment and benefit take up analysis was repeated using the 2024 Puerto Rico Community Survey Public Use Microdata Sample. For the results of this analysis, see Tables 17-19 in the Appendix.

Program Eligibility

Using the 2023 Puerto Rico Community Survey Public Use Microdata Sample data, program eligibility for NAP and SNAP was calculated based on household income and size as prescribed in the respective benefit schedules. Both NAP and SNAP define individual program eligibility as whether household income is below the maximum income cutoff for the given household size. Table 1 illustrates the monthly income cutoff and number of eligible individuals by household size for both programs. Overall, a total of 10696 individuals are eligible for SNAP and 8003 individuals are eligible for NAP out of a sample of 27247. This translates to 38.9% of the sample being eligible for SNAP and 29.18% for NAP.

Table 1: Income Conditions and Count of Eligible Individuals

	NAP		SNAP		
Household Size	Max Income	Number Eligible	Max Income	Number Eligible	Eligibility Change
1	619	2172	1215	3221	48.30%
2	1179	1984	1644	3016	52.02%
3	1706	1716	2072	2067	20.45%
4	2206	1196	2500	1380	15.38%
5	2666	625	2929	690	10.40%
6	3199	192	3357	204	6.25%
7	3599	63	3785	63	0
8	4112	16	4214	16	0
9	4632	18	4643	18	0
10	5152	10	5072	10	0
11	5666	11	5501	11	0
12	6186	0	5930	0	0
13	6699	0	6359	0	0
14	7219	0	6788	0	0
15	7732	0	7217	0	0
16	8252	0	7646	0	0
17	8766	0	8075	0	0
18	9286	0	8504	0	0
Total		8003		10696	33.65%

Further, the number of individuals receiving NAP benefits has seen an overall decrease from 2022 to 2023, though some categories of household size have experienced an increase. Table 2 displays the number of individuals that received NAP benefits and the respective percentage of the total sample broken down by household size. It is important to note that the eligibility criteria (maximum income cutoff) for NAP was the same in 2022 and 2023.

Table 2: NAP Receipt in 2022 and 2023

	2022		2023		
Household Size	Number Receiving	Percent Receiving	Number Receiving	Percent Receiving	Percent Change
1	2205	42.52%	2155	41.70%	−0.82%
2	3928	43.73%	3730	41.96%	−1.77%
3	3645	53.17%	3279	51.39%	−1.78%
4	2792	54.07%	2116	47.53%	−6.54%
5	1205	62.11%	1045	59.04%	−3.07%
6	330	63.22%	360	74.07%	10.85%
7	105	62.5%	140	86.96%	24.46%
8	56	100%	32	66.67%	−33.33%
9	27	100%	9	33.33%	−66.67%
10	10	100%	10	100%	0%
11	22	100%	11	50%	−50%
12	12	100%	12	100%	0%
Total		49.53%		47.03%	−2.5%

Program Benefits

Given that the PRCS PUMS dataset doesn't provide the actual value of benefits received by an individual and the fact that SNAP has never been implemented in Puerto Rico, benefit amounts for both programs were simulated using the respective 2023 program eligibility conditions and benefit schedules.

The program benefit schedules prescribe the maximum amount of benefits a household can receive each month, given their household size. Thus, yearly benefits amounts were calculated using the following formula:

$$\begin{aligned} \text{SimulatedBenefits} = & 12 \times \max(0, (\text{MaxBenefit}[\text{HHSize}] - (0.3 \times \text{HHIncome}/12)) \\ & + \text{ElderlyBonus} \end{aligned}$$

where the $\text{MaxBenefit}[\text{HHSize}]$ is the maximum benefit amount prescribed by the program schedule for that household size, HHIncome is the household's yearly income, and $\text{ElderlyBonus} = 1000$ if the household has a member that is 65 years or older (otherwise $\text{ElderlyBonus} = 0$).

Given that using household income directly in the calculation of simulated benefits presents endogeneity, a proxy for household income was used. This Avg_HHIncome was calculated by via a two step process: 1) for each individual in the data, averaged wage income was obtained by taking the average earnings in the region for individuals working in that occupation and of that gender, and 2) these average individual wage incomes were summed across households to get average household earnings. Thus, the benefit formula was adjusted to be:

$$\begin{aligned} \text{SimulatedBenefits} = & 12 \times \max(0, (\text{MaxBenefit}[\text{HHSize}] - (0.3 \times \text{Avg_HHIncome}/12)) \\ & + \text{ElderlyBonus} \end{aligned}$$

As a robustness check, the the study was run with both Avg_HHIncome and HHIncome and the results were compared. All but one of the regression coefficients were relatively similar in sign and magnitude and thus, Avg_HHIncome was taken to be a good proxy for household income. For the specific coefficient results, see Tables 11-16 of the Appendix.

Table 3 displays the maximum benefit schedules for NAP and SNAP, as well as the average simulated benefit amounts across household sizes for both programs. It is evident that, for every household size, SNAP provides larger average benefits than NAP. Further, overall average benefits are \$1,337 larger under SNAP.

Table 3: Benefit Schedules and Average Simulated Benefits

	NAP		SNAP		
Household Size	Max Benefit	Average Benefits	Max Benefit	Average Benefits	Percent Change
1	183	\$1860	291	\$2767	59.8%
2	337	\$2137	535	\$3431	61.6%
3	482	\$1344	766	\$2639	66.3%
4	612	\$1077	973	\$2421	69.2%
5	727	\$1747	1155	\$3945	69.3%
6	873	\$2067	1386	\$4764	69.7%
7	965	\$1943	1532	\$5092	72.4%
8	1103	\$2537	1751	\$6079	70.6%
9	1241	\$4864	1970	\$10529	68.4%
10	1379	\$971	2189	\$8769	90.0%
11	1517	\$486	2408	\$5452	91.8%
12	1655	0	2627	0	0
13	1792	0	2846	0	0
14	1930	0	3065	0	0
15	2068	0	3284	0	0
16	2206	0	3503	0	0
17	2344	0	3722	0	0
18	2482	0	3941	0	0
Overall		\$1701		\$3038	64.1%

Employment Response

Separate probit regressions were estimated for men and women, given that employment decisions vary by gender. Both of the employment regressions were weighted using the survey weights for each individual, $Weight_i$. Variable specifications are indicated in Table 4.

For Men:

$$\begin{aligned}
 Employment = \Phi(&\alpha + \beta_1 SimBenefit + \beta_2 EDU2 + \beta_3 EDU3 + \beta_4 EDU4 + \beta_5 EDU5 \\
 &+ \beta_6 EDU2 \times BLW21 + \beta_7 SimBenefit \times EDU2 + \beta_8 SimBenefit \times EDU3 \\
 &+ \beta_9 SimBenefit \times EDU4 + \beta_{10} SimBenefit \times EDUC5 + \beta_{11} Unemp \\
 &+ \beta_{12} Age + \beta_{13} Age^2 + \beta_{14} Married)
 \end{aligned}$$

For Women:

$$\begin{aligned}
 Employment = \Phi(&\alpha + \beta_1 SimBenefit + \beta_2 EDU2 + \beta_3 EDU3 + \beta_4 EDU4 + \beta_5 EDU5 \\
 &+ \beta_6 EDU2 \times BLW21 + \beta_7 SimBenefit \times EDU2 + \beta_8 SimBenefit \times EDU3 \\
 &+ \beta_9 SimBenefit \times EDU4 + \beta_{10} SimBenefit \times EDUC5 + \beta_{11} Unemp \\
 &+ \beta_{12} Age + \beta_{13} Age^2 + \beta_{14} Married + \beta_{15} NOC + \beta_{16} OCUnder6 + \beta_{17} FER \\
 &\beta_{18} NOC \times Married + \beta_{19} OCUnder6 \times Married + \beta_{20} FER \times Married)
 \end{aligned}$$

Table 4: Employment Regression Variables

Variable	Definition
<i>Employment</i>	1 if individual is employed, 0 otherwise.
<i>SimBenefit</i>	Benefit amount that an individual receives. Calculated by the previously described method. Baseline case is NAP, counterfactual is SNAP.
<i>EDU1</i>	1 if individual's highest received education is below high school, 0 otherwise. Base case of the <i>EDU</i> variables.
<i>EDU2</i>	1 if individual's highest received education is high school or GED diploma, 0 otherwise.
<i>EDU3</i>	1 if individual's highest received education is less than or more than 1 year of college without receiving a degree or an Associate's degree, 0 otherwise.

Variable	Definition
$EDU4$	1 if individual's highest received education is a Bachelor's degree, 0 otherwise.
$EDU5$	1 if individual's highest received education is Master's degree, professional, or Doctorate degree, 0 otherwise.
$EDU2 \times BLW21$	Interaction between $EDU2$ and the binary indicator of whether an individual is younger than 21.
$SimBenefit \times EDU2$	Interaction between $SimBenefit$ and $EDU2$.
$SimBenefit \times EDU3$	Interaction between $SimBenefit$ and $EDU3$.
$SimBenefit \times EDU4$	Interaction between $SimBenefit$ and $EDU4$.
$SimBenefit \times EDU5$	Interaction between $SimBenefit$ and $EDU5$.
$Unemp$	Unemployment rate in individual's PUMA region in 2023.
Age	Age of the individual.
Age^2	Squared age of the individual.
$Married$	1 if individual is married, 0 otherwise.
NOC	Number of own children an individual has.
$OCUnder6$	1 if individual has own children younger than 6 years old, 0 otherwise.
FER	1 if individual gave birth in past 12 months, 0 otherwise.
$NOC \times Married$	Interaction between NOC and $Married$.
$OCUnder6 \times Married$	Interaction between $OCUnder6$ and $Married$.
$FER \times Married$	Interaction between FER and $Married$.

Using the results of these two regressions, predicted values for employment were fitted for each individual to create $Employment_{NAP}$. Then a counterfactual was constructed, where the $SimBenefits$ calculated under the NAP benefit schedule were replaced with the $SimBenefits$ calculated under the SNAP benefit schedule. The regression results that were obtained for the baseline NAP program were used to fit predicted values of employment for each individual under this counterfactual, $Employment_{SNAP}$. $Employment_{NAP}$ and $Employment_{SNAP}$ were compared, for both men and women together, to determine

the change in employment across programs as follows:

$$\begin{aligned}BaselineEmployment &= \sum_i Employment_{NAP} \times Weight_i \\ EmploymentChange &= Employment_{SNAP} - Employment_{NAP} \\ EmploymentTotalChange &= \sum_i EmploymentChange_i \times Weight_i \\ Employment\%Change &= (EmploymentTotalChange / BaselineEmployment) * 100\end{aligned}$$

For more context on the employment behavior, the employment elasticity with respect to a change in benefits was calculated for each individual as follows:

$$\varepsilon_{Employment} = \frac{\phi(x_i) \times \beta_1 \times SimBenefit}{\Phi(x_i)}$$

where x_i is the probit regression for men's *Employment* if the individual is male or the probit regression for women's *Employment* if the individual is female.

Hours Worked Response

Separate regressions were estimated for men and women, given that labor force participation varies by gender. Both of the hours worked regressions were weighted using the designated survey weights for each individual, $Weight_i$. Variable specifications are indicated in Table 5.

For men:

$$\begin{aligned}
 WKHP = & \alpha + \beta_1 SimBenefit + \beta_2 WageIncome + \beta_3 EDU2 + \beta_4 EDU3 + \beta_5 EDU4 + \beta_6 EDU5 \\
 & + \beta_7 EDU2 \times BLW21 + \beta_8 SimBenefits \times EDU2 + \beta_9 SimBenefits \times EDU3 \\
 & + \beta_{10} SimBenefits \times EDU4 + \beta_{11} SimBenefits \times EDUC5 + \beta_{12} Age + \beta_{13} Age^2 \\
 & + \beta_{14} IND2 + \beta_{15} IND3 + \beta_{16} IND4 + \beta_{17} IND5 + \beta_{18} IND6 + \beta_{19} Married
 \end{aligned}$$

For women:

$$\begin{aligned}
 WKHP = & \alpha + \beta_1 SimBenefit + \beta_2 WageIncome + \beta_3 EDU2 + \beta_4 EDU3 + \beta_5 EDU4 + \beta_6 EDU5 \\
 & + \beta_7 EDU2 \times BLW21 + \beta_8 SimBenefits \times EDU2 + \beta_9 SimBenefits \times EDU3 \\
 & + \beta_{10} SimBenefits \times EDU4 + \beta_{11} SimBenefits \times EDUC5 + \beta_{12} Age + \beta_{13} Age^2 \\
 & + \beta_{14} IND2 + \beta_{15} IND3 + \beta_{16} IND4 + \beta_{17} IND5 + \beta_{18} IND6 + \beta_{19} Married \\
 & + \beta_{20} NOC + \beta_{21} OCUnder6 + \beta_{22} FER + \beta_{23} NOC \times Married \\
 & + \beta_{24} FER \times Married
 \end{aligned}$$

Table 5: Hours Worked Regression Variables

Variable	Definition
$WKHP$	Average number of hours an individual worked per week.
$SimBenefit$	Benefit amount that an individual receives. Calculated by the previously described method. Baseline case is NAP, counterfactual is SNAP.
$WageIncome$	Individual's wage income in past 12 months, adjusted to constant dollars.
$EDU1$	1 if individual's highest received education is below high school, 0 otherwise. Base case of the EDU variables.
$EDU2$	1 if individual's highest received education is high school or GED diploma, 0 otherwise.

Variable	Definition
<i>EDU3</i>	1 if individual's highest received education is less than or more than 1 year of college without receiving a degree or an Associate's degree, 0 otherwise.
<i>EDU4</i>	1 if individual's highest received education is a Bachelor's degree, 0 otherwise.
<i>EDU5</i>	1 if individual's highest received education is Master's degree, professional, or Doctorate degree, 0 otherwise.
<i>EDU2</i> $\times$ <i>BLW21</i>	Interaction between <i>EDU2</i> and the binary indicator of whether an individual is younger than 21.
<i>SimBenefit</i> $\times$ <i>EDU2</i>	Interaction between <i>SimBenefit</i> and <i>EDU2</i> .
<i>SimBenefit</i> $\times$ <i>EDU3</i>	Interaction between <i>SimBenefit</i> and <i>EDU3</i> .
<i>SimBenefit</i> $\times$ <i>EDU4</i>	Interaction between <i>SimBenefit</i> and <i>EDU4</i> .
<i>SimBenefit</i> $\times$ <i>EDU5</i>	Interaction between <i>SimBenefit</i> and <i>EDU5</i> .
<i>Age</i>	Age of the individual.
<i>Age</i> ²	Squared age of the individual.
<i>IND1</i>	1 if individual worked in Agriculture, Construction, Mining, or Utilities industry. Base case of the <i>IND</i> variables.
<i>IND2</i>	1 if individual worked in Manufacturing industry.
<i>IND3</i>	1 if individual worked in Wholesale, Retail, or Transportation industry.
<i>IND4</i>	1 if individual worked in Information or Finance industry.
<i>IND5</i>	1 if individual worked in Professional, Administration, or Services industry.
<i>IND6</i>	1 if individual worked in Education, Medical, Social Assistance, Entertainment, or Military industry.
<i>Married</i>	1 if individual is married, 0 otherwise.
<i>NOC</i>	Number of own children an individual has.
<i>OCUnder6</i>	1 if individual has own children younger than 6 years old, 0 otherwise.
<i>FER</i>	1 if individual gave birth in past 12 months, 0 otherwise.
<i>NOC</i> $\times$ <i>Married</i>	Interaction between <i>NOC</i> and <i>Married</i> .
<i>FER</i> $\times$ <i>Married</i>	Interaction between <i>FER</i> and <i>Married</i> .

Using the results of this regression, predicted values for number of hours worked ($WKHP$) were fitted for each individual to create $WKHP_{NAP}$. Then a counterfactual was constructed, where $SimBenefits$ calculated under the NAP benefit schedule was replaced with the $SimBenefits$ calculated under the SNAP benefit schedule. The regression results that were obtained for the baseline NAP program were used to fit predicted values of number of hours worked for each individual under this counterfactual, $WKHP_{SNAP}$. $WKHP_{NAP}$ and $WKHP_{SNAP}$ were compared to determine the change in number of hours worked across programs as follows:

$$\begin{aligned}
BaselineHours &= \sum_i WKHP_{NAP} \times Weight_i \\
HoursChange &= WKHP_{SNAP} - WKHP_{NAP} \\
HoursTotalChange &= \sum_i HoursChange_i \times Weight_i \\
Hours\%Change &= (HoursTotalChange / BaselineHours) * 100
\end{aligned}$$

For more context on the individual impacts of benefits on weekly hours worked, the elasticity of weekly hours worked with respect to a change in benefit amounts was calculated for each individual as follows:

$$\varepsilon_{WKHP} = \beta_1 * \left(\frac{SimBenefit}{WKHP} \right)$$

Benefit Take Up Response

Separate probit regressions were estimated for men and women. Both of these regressions were weighted using the designed survey weights for each individual, $Weight_i$. Variable specifications are indicated in Table 6.

For men:

$$\begin{aligned}
P(WFReciept|Elig = 1) = & \Phi(\alpha + \beta_1 SimBenefit + \beta_2 HHSize + \beta_3 R65 + \beta_4 EDU2 \\
& + \beta_5 EDU3 + \beta_6 EDU4 + \beta_7 EDU5 + \beta_8 SimBenefits \times EDU2 \\
& + \beta_9 SimBenefits \times EDU3 + \beta_{10} SimBenefits \times EDU4 \\
& + \beta_{11} SimBenefits \times EDU5 + \beta_{12} Age + \beta_{13} Age^2 + \beta_{14} Married \\
& + \beta_{15} FS_Availability + \beta_{16} FS_LagReceipt)
\end{aligned}$$

For women:

$$\begin{aligned}
P(WFReciept|Elig = 1) = & \Phi(\alpha + \beta_1 SimBenefit + \beta_2 HHSize + \beta_3 R65 + \beta_4 EDU2 \\
& + \beta_5 EDU3 + \beta_6 EDU4 + \beta_7 EDU5 + \beta_8 SimBenefits \times EDU2 \\
& + \beta_9 SimBenefits \times EDU3 + \beta_{10} SimBenefits \times EDU4 \\
& + \beta_{11} SimBenefits \times EDU5 + \beta_{12} Age + \beta_{13} Age^2 + \beta_{14} Married \\
& + \beta_{15} FS_Availability + \beta_{16} FS_LagReceipt + \beta_{16} NOC \\
& + \beta_{17} OCUnder6 + \beta_{18} FER + \beta_{19} NOC \times Married \\
& + \beta_{20} OCUnder6 \times Married + \beta_{21} FER \times Married)
\end{aligned}$$

Table 6: Benefit Take Up Regression Variables

Variable	Definition
<i>Elig</i>	1 if individual is eligible for the welfare program as determined by the program schedule, 0 otherwise.
<i>WFReceipt</i>	1 if the individual received NAP benefits, 0 otherwise.
<i>SimBenefit</i>	Benefit amount that an individual receives. Calculated by the previously described method. Baseline case is NAP, counterfactual is SNAP.
<i>HHSize</i>	Number of people in an individual's household.

Variable	Definition
<i>R65</i>	1 if individual's household has a member aged 65 or older, 0 otherwise.
<i>EDU1</i>	1 if individual's highest received education is below high school, 0 otherwise. Base case of the <i>EDU</i> variables.
<i>EDU2</i>	1 if individual's highest received education is high school or GED diploma, 0 otherwise.
<i>EDU3</i>	1 if individual's highest received education is less than or more than 1 year of college without receiving a degree or an Associate's degree, 0 otherwise.
<i>EDU4</i>	1 if individual's highest received education is a Bachelor's degree, 0 otherwise.
<i>EDU5</i>	1 if individual's highest received education is Master's degree, professional, or Doctorate degree, 0 otherwise.
<i>SimBenefit</i> $\times$ <i>EDU2</i>	Interaction between <i>SimBenefit</i> and <i>EDU2</i> .
<i>SimBenefit</i> $\times$ <i>EDU3</i>	Interaction between <i>SimBenefit</i> and <i>EDU3</i> .
<i>SimBenefit</i> $\times$ <i>EDU4</i>	Interaction between <i>SimBenefit</i> and <i>EDU4</i> .
<i>SimBenefit</i> $\times$ <i>EDU5</i>	Interaction between <i>SimBenefit</i> and <i>EDU5</i> .
<i>Age</i>	Age of the individual.
<i>Age</i> ²	Squared age of the individual.
<i>Married</i>	1 if individual is married, 0 otherwise.
<i>FS_Availability</i>	Ratio of number of businesses authorized to accept NAP food stamps in the individual's PUMA region over the population count in the individual's PUMA region in 2023.
<i>FS_LagReceipt</i>	Number of individuals that received NAP food stamps in the individual's PUMA region in 2022.
<i>NOC</i>	Number of own children an individual has.
<i>OCUnder6</i>	1 if individual has own children younger than 6 years old, 0 otherwise.
<i>FER</i>	1 if individual gave birth in past 12 months, 0 otherwise.
<i>NOC</i> $\times$ <i>Married</i>	Interaction between <i>NOC</i> and <i>Married</i> .
<i>OCUnder6</i> $\times$ <i>Married</i>	Interaction between <i>OCUnder6</i> and <i>Married</i> .
<i>FER</i> $\times$ <i>Married</i>	Interaction between <i>FER</i> and <i>Married</i> .

Using the results of this regressions, predicted values for welfare receipt were fitted for each individual to create $Receipt_{NAP}$. Then a counterfactual was constructed, where the $SimBenefits$ calculated under the NAP benefit schedule were replaced with the $SimBenefits$ calculated under the SNAP benefit schedule. The regression results that were obtained for the baseline NAP program were used to fit predicted values of welfare receipt for each individual under this counterfactual, $Receipt_{SNAP}$. $Receipt_{NAP}$ and $Receipt_{SNAP}$ were compared to determine the change in welfare take up across programs as follows:

$$\begin{aligned}
BaselineTakeup &= \sum_i Receipt_{NAP} \times Weight_i \\
TakeupChange &= Receipt_{SNAP} - Receipt_{NAP} \\
TakeupTotalChange &= \sum_i TakeupChange_i \times Weight_i \\
Takeup\%Change &= (TakeupTotalChange / BaselineTakeup) * 100
\end{aligned}$$

For more context on welfare receipt behavior, the take up elasticity with respect to a change in benefits was calculated for each individual as follows:

$$\varepsilon_{Takeup} = \frac{\phi(x_i) \times \beta_1 \times SimBenefit}{\Phi(x_i)}$$

where x_i is the probit regression for men's $WFReceipt|Elig = 1$ if the individual is male or the probit regression for women's $WFReceipt|Elig = 1$ if the individual is female.

Results

Employment Response

Coefficients and average marginal effects for the men's and women's employment regressions are displayed in Table 7. These results are scaled by 100 for easier readability.

For men, all coefficients are significant at the 1% level. For women, all coefficients are significant at the 1% level, with the exception of *NOC* being significant at the 5% level and *SimBen* $\times$ *EDU5* not being significant. The coefficient on *SimBenefit* is -0.0200 for men and -0.0300 for women. This aligns with the economic theory that increasing the amount of welfare benefits will decrease the likelihood of an individual's employment. Further, all *SimBen* $\times$ *EDU* coefficients are negative for men, while these same coefficients are positive for women (with the exception of *SimBen* $\times$ *EDU4* being negative for women). This implies that, for men, the negative effect of welfare benefits on employment probability is larger for individuals with higher than a high school education and, in particular, having an education higher than a Bachelor's degree causes the largest amplification of the negative effect of food stamp benefits. Women see a generally opposite effect, where the negative effect of welfare benefits on employment probability is smaller for individuals with higher than a high school education.

The controls for education, age, and marriage status all show reasonable signs and magnitude for men and women. A higher level of completed education increasingly raises the probability of employment, especially for women. Employment increases with age but at a decreasing rate and this increase is stronger for men. Marriage increases the probability of employment, with a larger impact seen for men. Further, a higher unemployment rate in the individual's region decreases the probability of employment for men, while increasing it for women, though at a relatively smaller magnitude. For women, a greater number of children and recently giving birth reduces the employment probability, while married women are more likely to be employed given these same characteristics. However, a greater number of children under 6 years old corresponds to higher probability of employment for women with women that are married having a reduction in probability of employment under these circumstances.

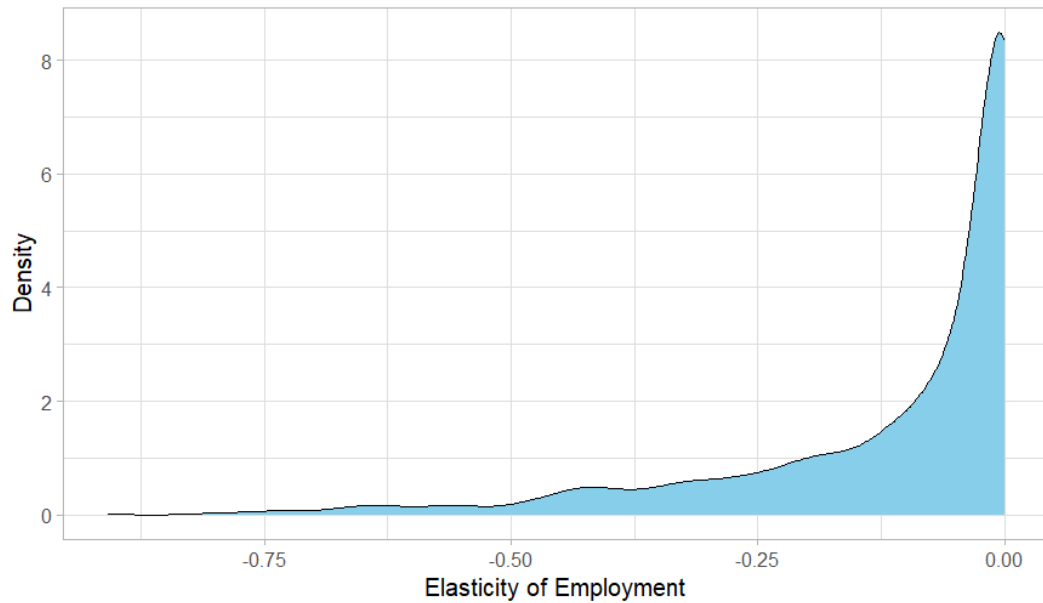
Table 7: Employment Regression Coefficients for Men and Women (Scaled by 100)

	Men		Women	
	<i>Employment</i>	<i>AME</i>	<i>Employment</i>	<i>AME</i>
<i>SimBenefit</i>	−0.0200 *** (0.0004)	−0.0022	−0.0300 *** (0.0007)	−0.0037
<i>EDU2</i>	10.340 *** (0.900)	1.3600	38.660 *** (1.400)	5.0500
<i>EDU3</i>	21.600 *** (0.900)	2.8300	58.090 *** (1.300)	7.5900
<i>EDU4</i>	36.480 *** (1.000)	4.7800	85.750 *** (1.400)	11.200
<i>EDU5</i>	45.610 *** (1.400)	5.9800	79.410 *** (1.500)	10.370
<i>EDU2 × BLW2</i>	−13.510 *** (1.500)	−1.7700	−37.760 *** (1.600)	−4.9300
<i>SimBen × EDU2</i>	−0.0072 *** (0.0005)	−0.0009	0.0028 *** (0.0007)	0.0004
<i>SimBen × EDU3</i>	−0.0200 *** (0.0005)	−0.0020	0.0056 *** (0.0007)	0.0007
<i>SimBen × EDU4</i>	−0.0100 *** (0.0005)	−0.0016	−0.0020 *** (0.0008)	−0.0003
<i>SimBen × EDU5</i>	−0.0200 *** (0.0013)	−0.0027	0.0022 (0.0013)	0.0003
<i>Unemp</i>	−5.7900 *** (0.100)	−0.7600	2.9800 *** (0.100)	0.3900
<i>Age</i>	6.7000 *** (0.100)	0.8800	3.2400 *** (0.100)	0.4200
<i>Age²</i>	−0.0700 *** (0.0010)	−0.0088	−0.0200 *** (0.0013)	−0.0029
<i>Married</i>	36.790 *** (0.600)	4.8200	19.600 *** (0.700)	2.5600
<i>NOC</i>	—	—	−0.8700 ** (0.400)	−0.1100
No. obs.	5048		4895	

	Men		Women	
	<i>Employment</i>	<i>AME</i>	<i>Employment</i>	<i>AME</i>
<i>OCUnder6</i>	—	—	10.390 *** (1.100)	1.3600
<i>FER</i>	—	—	−7.7600 *** (2.000)	−1.0100
<i>NOC</i> × <i>Married</i>	—	—	9.0700 *** (0.800)	1.1180
<i>OCUnder6</i> × <i>Married</i>	—	—	−15.540 *** (2.200)	−2.0300
<i>FER</i> × <i>Married</i>	—	—	42.000 *** (2.000)	5.4800
<i>CNST</i>	7.8500 *** (2.200)	—	−22.760 *** (2.500)	—
No. obs.	5048		4895	

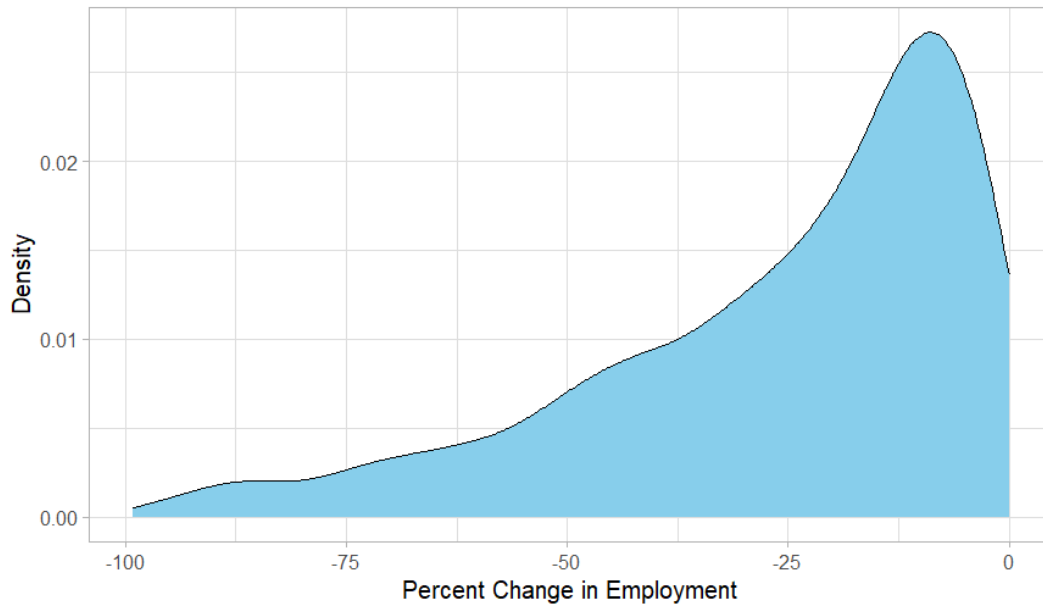
Given the similarity in the coefficient estimates for *SimBenefit* across men and women (-0.02 vs -0.03), the elasticity of employment in response to a change in simulated benefit amount was obtained for both men and women combined. The distribution of this elasticity is depicted in Figure 6. The employment elasticity signifies the percentage change the employment probability experiences in response to a 1% change in the benefit amounts. The average employment elasticity in the sample was found to be -0.02518 and all elasticities are negative. This aligns with the argument that high benefit amounts will disincentivize employment since this increases the benefits of being unemployed and receiving welfare payments instead of being employed. The majority of elasticity values falling above -0.25 and being especially close to 0 suggests that although the impact of benefits on employment is negative, it may not be highly strong of an impact. Given that the average simulated benefits were \$1,701 under NAP and \$3,038 under SNAP, this represents a 64.1% increase in average benefits. Thus, this increase in benefits translates to a 1.614% relative decrease in individual employment probability on average, which is relatively small for each individual, but may represent a larger impact on the population level.

Figure 6. Distribution of Individual Elasticity of Employment



Further, the distribution of the changes in individual's employment is illustrated in Figure 7. The average change in an individual's employment was found to be -5.95528 percentage points and all observed changes are either negative or zero. Further, calculating the overall change in employment for the sample it is found that employment reduces by 4.60923% . When expanding the sample results to the entire population of Puerto Rico (via the survey weights), this corresponds to $54,926$ less individuals being employed under SNAP than under NAP. Given that the baseline under NAP had $1,191,632$ employed individuals and $119,291$ unemployed individuals, the counterfactual scenario under SNAP will have $1,136,706$ employed individuals and $174,217$ unemployed individuals. This represents a change from an unemployment rate of 9.0998% under NAP to an unemployment rate of 13.2896% under SNAP, which is a 4.1898% increase in overall unemployment rate in Puerto Rico.

Figure 7. Distribution of Individual Change in Employment Over Programs



Hours Worked Response

Coefficients for the men's and women's employment regressions are displayed in Table 8. For men, all coefficients except *IND3*, *IND4*, and *IND5* are significant at at least the 1% level. For women, all coefficients except *EDU2*, *SimBen* $\times$ *EDU5*, *IND2*, *IND4*, *IND5*, *IND6*, *Married*, and all women specific variables are significant at at least the 1% level. The coefficient on *SimBenefit* is -0.0016 for men and 0.0012 for women. This suggests that for men, an increase in food stamp benefits reduces the number of weekly hours worked as is hypothesized, while, for women, this change in benefits actually increases the number of weekly hours worked. Further, all *SimBen* $\times$ *EDU* coefficients are positive for men, while these same coefficients are negative for women. This implies that, for men, the negative effect of welfare benefits on hours worked is smaller for individuals with higher than a high school education. For women, given that the coefficient on *SimBenefit* was positive, the positive effect of welfare benefits on hours worked is larger for individuals with higher than a high school education.

The controls for education and age both show intuitive signs and magnitudes. A higher level of completed education increases the number of weekly hours worked, especially for women. The number of hours worked increases with age but at a decreasing rate, with the increase being stronger for men. For men, only *IND2* and *IND6* are significant where working in *IND2* increases and working in *IND6* decreases the weekly number of hours worked as compared to the base case of *IND1*. Given that *IND1* is agriculture, construction, mining, or utilities it is reasonable that working in manufacturing (*IND2*) may be more demanding and would require more weekly hours and that working in education, medical, social assistance, entertainment, or military (*IND6*) would require less weekly hours. For women, only *IND3* is significant and it is negative. By similar logic, it is also reasonable that working in wholesale, retail, or transportation (*IND3*) is less demanding and requires less weekly hours than *IND1*. It is interesting that the women specific variables are not significant in the women's regression, however it can be argued that having children/the number of children mainly impacts the overall decision of whether to work or not, rather than how many hours to work. The fact that the women specific variables were significant in the employment regression supports this thought.

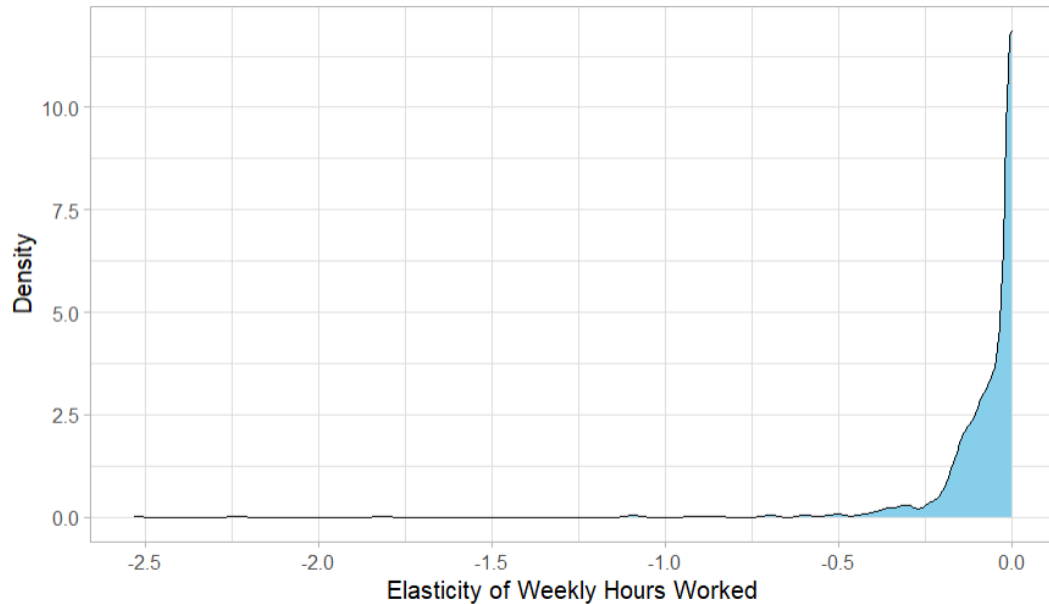
Table 8: Hours Worked Regression Coefficients for Men and Women

	Men	Women
	<i>WKHP</i>	<i>WKHP</i>
<i>SimBenefit</i>	−0.0016 *** (0.0000)	0.0012 * (0.0010)
<i>WageIncome</i>	0.00005 *** (0.000004)	0.00008 *** (0.000005)
<i>EDU2</i>	1.4622 ** (0.5860)	1.3886 (1.0330)
<i>EDU3</i>	1.3909 ** (0.5910)	3.5685 *** (1.0030)
<i>EDU4</i>	2.2870 *** (0.6180)	5.1185 *** (1.0050)
<i>EDU5</i>	2.6342 *** (0.7410)	5.0457 *** (1.0480)
<i>EDU2</i> × <i>BLW2</i>	−2.6435 ** (1.2470)	−4.1188 ** (1.3300)
<i>SimBen</i> × <i>EDU2</i>	0.0011 ** (0.0000)	−0.0021 *** (0.0010)
<i>SimBen</i> × <i>EDU3</i>	0.0011 ** (0.0010)	−0.0025 *** (0.0010)
<i>SimBen</i> × <i>EDU4</i>	0.0027 *** (0.0010)	−0.0022 *** (0.0010)
<i>SimBen</i> × <i>EDU5</i>	0.0023 * (0.0010)	−0.0001 (0.0010)
<i>Age</i>	0.7488 *** (0.0600)	0.6280 *** (0.0680)
<i>Age</i> ²	−0.0083 *** (0.0010)	−0.0070 *** (0.0010)
<i>IND2</i>	1.3160 ** (0.5780)	1.6212 (1.1190)
<i>IND3</i>	0.2236 (0.4930)	−2.5285 ** (1.0550)
<i>IND4</i>	0.7950 (0.6830)	0.4924 (1.1040)
No. obs.	4963	4812

	Men	Women
	<i>WKHP</i>	<i>WKHP</i>
<i>IND5</i>	−0.5976 (0.4500)	−1.0524 (1.0320)
<i>IND6</i>	−1.4681 *** (0.4960)	−1.3166 (1.0150)
<i>Married</i>	0.7544 ** (0.3230)	−0.0888 (0.3770)
<i>NOC</i>	—	−0.3672 (0.2550)
<i>OCUnder6</i>	—	−0.2569 (0.5830)
<i>FER</i>	—	0.3185 (1.2690)
<i>NOC × Married</i>	—	0.1054 (0.3730)
<i>FER × Married</i>	—	−0.9793 (1.9460)
<i>CNST</i>	18.7576 *** (1.3970)	18.1637 *** (1.8790)
No. obs.	4963	4812

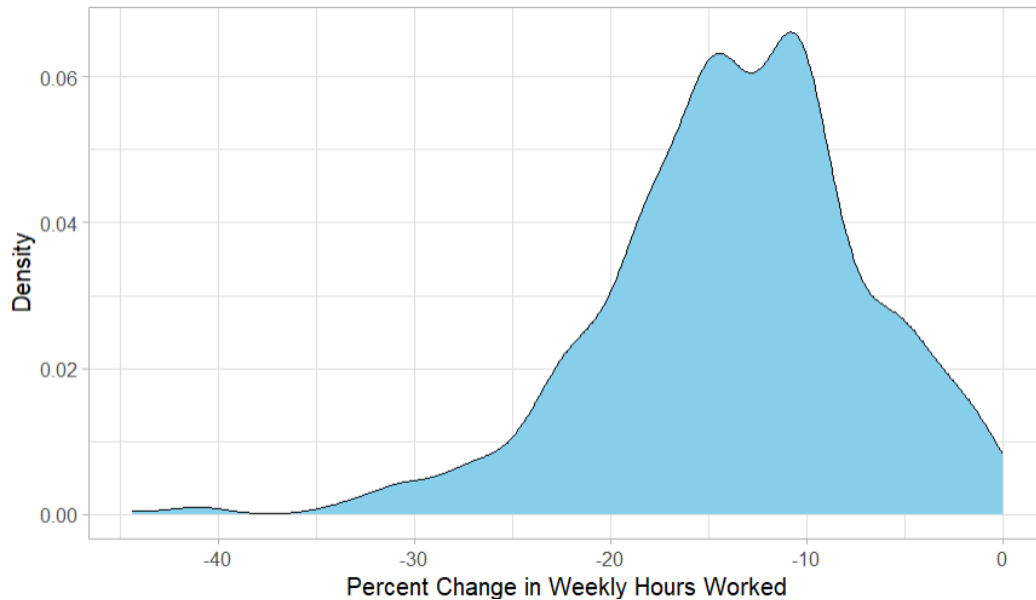
Based on the weekly hours worked regression results, the elasticity of weekly hours worked in response to a change in simulated benefit amount was derived for each individual in the sample. The weekly hours worked elasticity signifies the percentage change in the number of weekly hours worked in response to a 1% change in the benefit amount. The distribution of these elasticities is depicted in Figure 8. The average weekly hours worked elasticity was found to be -0.00043 and all observed elasticities were negative. This aligns with the results found for the impact of benefits on employment probability, where in this case, higher benefits incentivize decreasing the amount of time spent at work such that one can instead claim welfare benefits. Given the average simulated benefits of \$1,701 under NAP and \$3,038 under SNAP, a transition from NAP to SNAP would result in an average 0.02756% relative decrease in the number of hours an individual works. This is a relatively small change on the individual level.

Figure 8. Distribution of Individual Elasticity of Weekly Hours Work



The change in the number of weekly hours worked was further found for each individual. Figure 9 shows the distribution of the percent change in weekly hours worked after a transition from NAP to SNAP. The average change in weekly hours in the sample was found to be -0.24633% and all observed changes were negative. Further, expanding these sample specific results to the entire population of Puerto Rico using survey weights, the overall change in number of weekly hours worked when transitioning from NAP to SNAP was -0.33521% . This translates to a decrease of 154,426 weekly hours worked under SNAP. The baseline under NAP had a total of 46,067,707 weekly hours worked by the entire population of Puerto Rico (which translates to roughly 1,151,693 individuals working a 40 hour week). In contrast, the counterfactual under SNAP had a total of 45,913,281 weekly hours worked by the entire population of Puerto Rico (which translates to roughly 1,147,832 individuals working a 40 hour week).

Figure 9. Individual Change in Weekly Hours Worked Across Programs



Benefit Take Up Response

Coefficients and average marginal effects for the men's and women's benefit take up regressions are displayed in Table 9. These results are scaled by 100 for easier readability.

For men, all coefficients are significant at the 1% level. For women, all coefficients are significant at the 1% level, excluding $OCUnder6 \times Married$ which is significant at the 5%, and FER and $NOC \times Married$ which are not significant. The coefficient on $SimBenefit$ is 0.0035 for men and 0.0049 for women. This aligns with the theory that an increase in food stamps benefits should increase the number of recipients as there are higher incentives for eligible individuals to actually collect these benefits. Further, all $SimBen \times EDU$ coefficients are positive for men, except $SimBen \times EDU4$ which is negative. Conversely, all $SimBen \times EDU$ coefficients are negative for women, except $SimBen \times EDU5$ which is positive.

A higher level of completed education as well as marriage decreases the probability of receiving benefits for both men and women, which aligns with expectations. Benefit receipt probability increases with age but at a decreasing rate and this increase is stronger for women. Greater availability of food stamps in the region increases the probability of receiving benefits, while a greater number of people receiving food stamps in the region in 2022 decreases the probability of receiving benefits. This corresponds with the fact that accessibility is important for welfare receipt and that larger populations claiming benefits may reduce the availability of those benefits in that region in the future.

Having more children, especially those under the age of 6, increases the probability of a woman receiving benefits even when the woman is married, which is explainable by the high cost of raising children. The fact that giving recently giving birth reduces the probability of receiving benefits, especially if the woman is married, can potentially be explained by the fact that receiving benefits involves a time consuming process of visiting various government offices which is likely not feasible while having a newborn.

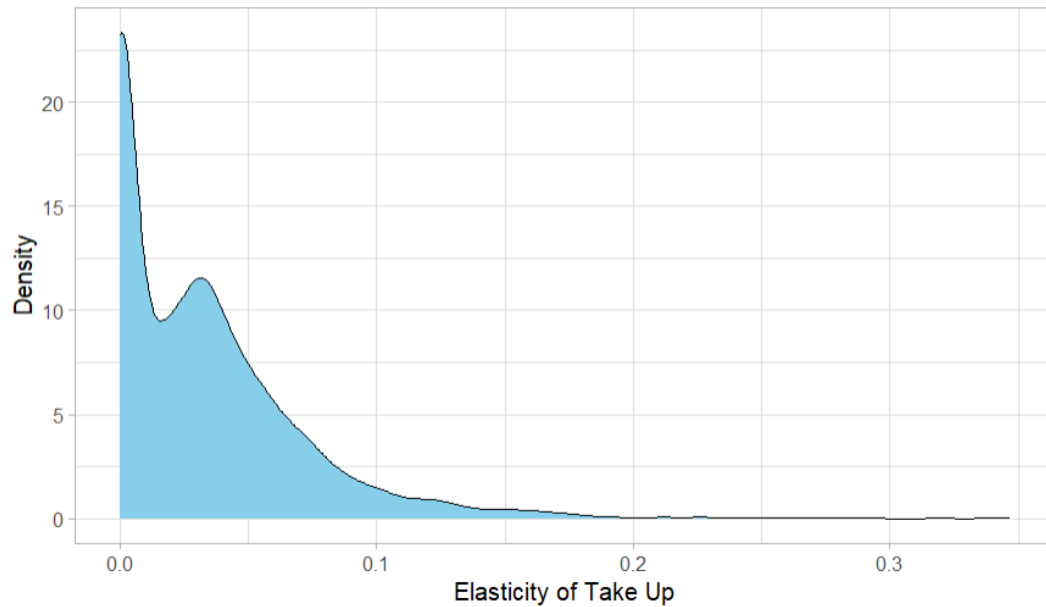
Table 9: Take up Rate Regression Coefficients for Men and Women (Scaled by 100)

	Men		Women	
	<i>Takeup</i>	<i>AME</i>	<i>Takeup</i>	<i>AME</i>
<i>SimBenefit</i>	0.0035 *** (0.0002)	0.0008	0.0049 *** (0.0002)	0.0011
<i>HHSize</i>	10.530 *** (0.200)	2.5500	5.2600 *** (0.200)	1.1200
<i>R65</i>	-19.100 *** (0.600)	-4.6200	-9.4800 *** (0.600)	-2.0100
<i>EDU2</i>	-15.720 *** (1.000)	-3.8100	-15.450 *** (1.100)	-3.2800
<i>EDU3</i>	-50.910 *** (1.100)	-12.330	-24.980 *** (1.100)	-5.3100
<i>EDU4</i>	-60.140 *** (1.700)	-14.560	-90.600 *** (1.300)	-19.250
<i>EDU5</i>	-122.27 *** (2.900)	-29.610	-151.65 *** (2.100)	-32.220
<i>SimBen</i> $\times$ <i>EDU2</i>	0.0008 *** (0.0003)	0.0002	-0.0044 *** (0.0003)	-0.0009
<i>SimBen</i> $\times$ <i>EDU3</i>	0.0010 *** (0.0003)	0.0002	-0.0066 *** (0.0003)	-0.0014
<i>SimBen</i> $\times$ <i>EDU4</i>	-0.0062 *** (0.0005)	-0.0015	-0.0025 *** (0.0004)	-0.0005
<i>SimBen</i> $\times$ <i>EDU5</i>	0.0100 *** (0.0010)	0.0024	0.0033 *** (0.0007)	0.0007
<i>Age</i>	0.2100 *** (0.100)	0.0500	3.7700 *** (0.100)	0.8000
<i>Age</i> ²	-0.0002 (0.0006)	-0.00004	-0.0400 *** (0.0006)	-0.0080
<i>Married</i>	-23.750 *** (0.700)	-5.7500	-43.160 *** (0.800)	-9.1700
<i>FS_Availability</i>	406.45 *** (10.100)	98.410	449.92 *** (9.300)	95.590
<i>FS_LagReceipt</i>	-0.0400 *** (0.0012)	-0.0086	-0.0300 *** (0.0011)	-0.0073
No. obs.	2910		3896	

	Men		Women	
	<i>Takeup</i>	<i>AME</i>	<i>Takeup</i>	<i>AME</i>
<i>NOC</i>	—	—	7.9300 *** (0.400)	1.6900
<i>OCUnder6</i>	—	—	46.530 *** (1.100)	9.8900
<i>FER</i>	—	—	−3.1600 (2.000)	−0.6700
<i>NOC</i> × <i>Married</i>	—	—	0.8900 (0.800)	0.1900
<i>OCUnder6</i> × <i>Married</i>	—	—	5.8400 ** (2.800)	1.2400
<i>FER</i> × <i>Married</i>	—	—	−74.970 *** (4.900)	−15.930
<i>CNST</i>	61.640 *** (1.700)	—	20.490 *** (1.700)	—
No. obs.	2910		3896	

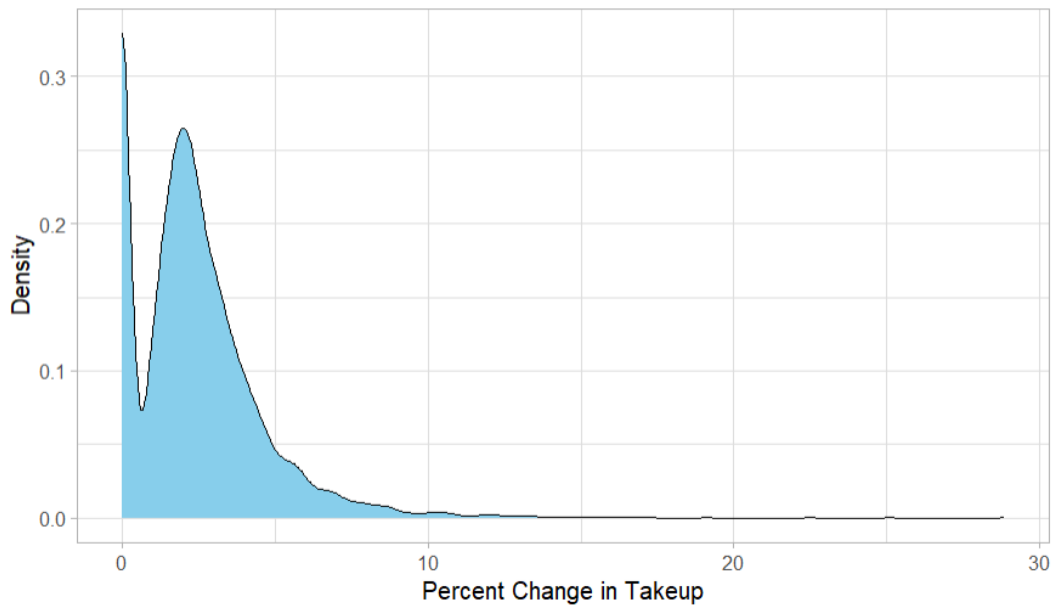
Based on the above regression results, the elasticity of take up in response to a change in simulated benefit amount was derived and its distribution is depicted in Figure 10. This elasticity signifies the percent change in benefit take up probability in response to a 1% change in the benefit amount. The average take up elasticity in the sample was found to be 0.03128 and all observed elasticities were positive. This aligns with the argument that higher benefit amounts would present greater incentive to collect these benefits. Given that the average simulated benefits increased from \$1,701 under NAP to \$3,038 under SNAP, this translates to a 2.005% relative increase in individual benefit take up probability on average.

Figure 10. Distribution of Individual Elasticity of Take Up



Further, the distribution of the percent change in take up for each individual when going from NAP to SNAP is displayed in Figure 11. The average change in take up in the sample was found to be 2.33911% and all observed changes were positive. Moreover, the change in overall benefit take up when transitioning to SNAP from NAP was found to be 2.24297%. When expanding the sample results to the entire population of Puerto Rico, this translates to 15,941 more individuals receiving benefits under SNAP than NAP given that they are eligible for the program. The baseline under NAP had 710,702 eligible individuals receive benefits and 133,135 eligible individuals not receive benefits, while the counterfactual scenario under SNAP has 726,643 eligible individuals receiving benefits and 117,194 eligible individuals not receiving them. This means NAP had a take up rate of 84.2227% while SNAP has one of 86.1118%, an increase of 1.8891%.

Figure 11. Distribution of Individual Change in Takeup Rate Over Programs



Potential Improvements

There are several factors that impact employment and welfare programs that could be taken into account to improve the results of this counterfactual simulation, with the most pressing ones being the informal labor market, food stamp program employment requirements, and price differences between the United States and Puerto Rico.

An important consideration is the informal labor market, which makes up a large percentage of labor activity in Puerto Rico, especially for men. Castro-González and Medina León (2017) concluded that the informal economy made up 36.47% of Puerto Rico's GDP in 2014. Although there have not been any follow up papers that directly quantify this value for the present day, it is likely that this number has sustained its magnitude, if not increased. Low employment rates for Puerto Rican men have also been extensively documented throughout history. As Enchautegui and Freeman (2005) put it, a potential explanation for this is the "Rich Uncle (Sam) Hypothesis" where the U.S.-funded transfer payments that make up a large percentage of family's income and unmeasured informal sector employment contribute to the officially documented low employment rates. This question of informal employment is particularly pertinent to the results of this paper, given that it is not possible to observe whether or not someone works informally solely based on the variables given in the Puerto Rico Community Survey data which was used for analysis. The questions of whether an individual that is official marked as unemployed is truly unemployed or is working informally and whether an individual that qualifies for food stamp benefits is truly earning less than the income cutoff or is not reporting informal income sources in order to still qualify are consequential for both accurately estimating eligibility counts & benefit amounts, and determining the impacts of a program transition on the labor market. Given the lack of access to data that reports on individuals true employment, both formally and informally, all analysis was run under the assumption that all reported data was true (i.e. unemployed individuals are truly unemployed and an individual's income is their true income). If this is not the case, the results of the analysis would likely overestimate the impacts of a transition to SNAP. Further policy analysis on the topic of food stamps would benefit from gathering outside data which captures the true relationship between the formal and informal labor markets and economic outcomes.

Further, there is a work requirement under SNAP. Per the SNAP guidelines, all able-bodied individuals that do not receive exemption must work or volunteer 80 hours per month. Since that this is not a full-time work commitment that individuals must make in order to receive benefits and the fact that the majority of the sample worked full time, this feature was not included in the model for employment. However, this requirement for at least some level of employment may help in mitigating the disincentives observed for SNAP benefits. However, this work requirement also falls into the previous question of informal labor activity given that many individuals that are already receiving benefits are likely to be working informally and thus a transition to formal work (or reporting their informal work) in order to continue receiving benefits, may not produce any positive benefits for this large group of people. In further analysis with information on the informal market, it would be interesting to estimate the impacts of this work requirement on reducing employment disincentives on both officially unemployed and unofficially employed individuals.

The difference in prices between the United States and Puerto Rico may also have impact the results of this paper. Given that average annual incomes for jobs in Puerto Rico tend to be lower as compared to that of their equivalents in the United States, the argument that it is cheaper to live in Puerto Rico arises. Table 12 presents the average prices of a selected variety of items in September 2025, with data for the United States from the United States Bureau of Labor Statistics and data for Puerto Rico from the Selina Wamuchi and Trading Economics price databases. Though rent and gasoline are significantly cheaper in Puerto Rico, the cost of groceries and other household items is either at the same level or more expensive than in the U.S.. Given this information, the analysis conducted in this paper was done with the assumption that price levels in the United States and Puerto Rico are the same and that the impact of a given amount of benefits would be comparable in both regions. Recent data from 2025 was used for Puerto Rico given that equivalent pricing data for 2023 is limited. It is possible that prices in 2023 were indeed varied enough to break the assumption that benefits would have comparable impacts for both Puerto Rico and the United States. In conjunction with the gathering of data on informal employment measures, it would be helpful to also collect information about household's monthly expenses, which when compared to that of the United States, would shed light into this question of whether differences in average prices between the two regions are significant.

Table 10: Prices of Select Items in United States and Puerto Rico

	United States	Puerto Rico
Item	Average Price	Average Price
Bread	\$1.84	\$2.76
Chicken (lb)	\$2.06	\$2.20
Eggs (dozen)	\$3.49	\$5.79
Milk (gal)	\$4.13	\$5.75
Oranges (lb)	\$1.80	\$0.41
Tomatoes (lb)	\$1.91	\$2.34
Electricity (kWh)	\$0.19	\$0.33
Gasoline (gal)	\$3.34	\$2.84
Rent (1 bdrm)	\$1,629	\$550

Conclusion

Employment elasticity with respect to a change in benefit amount is negative for all individuals, with an average value of -0.02518 . This aligns with the economic theory that increased benefits decrease the receiving population's willingness to work as the benefits to being unemployed become higher and potentially outweigh the benefits of employment. A transition from NAP to SNAP increases unemployment in the sample by 4.60923%, while expanding the results to the overall population of Puerto Rico yields a 4.1898% increase in the unemployment rate. This corresponds to 54,926 less individuals being employed under SNAP than under NAP.

The weekly hours worked elasticity with respect to a change in benefit amount is negative for all individuals, with an average value of -0.00043 . This confirms the finds of the employment analysis, where an increase in benefits decreases an individuals willingness to work, evident through both their employment probability and the number of hours worked weekly. A transition from NAP to SNAP decreases the number of hours worked in by individuals in the sample by 0.33521%. Expanding these sample results to all of Puerto Rico yields a decrease of 154,426 weekly hours worked by the entire population.

Take up rate elasticity, defined as the probability of an individual receiving benefits given that they are eligible, with respect to a change in benefit amount is positive for all individ-

uals. This elasticity was found to have an average value of 0.03128. This also aligns with the conclusion that higher benefit amounts provide greater encouragement for individuals to actually receive the assistance, given that there is oftentimes certain stigmas or time consuming processes that one must overcome to receive benefits. Further, a transition from NAP to SNAP increases the benefit take up in the sample by 2.24297%, while expanding the results to the overall population of Puerto Rico yields a 1.8891% increase in the benefit take up rate. This translates to 15,941 more eligible individuals actually receiving benefits under SNAP than under NAP.

These estimates of the impacts of a program change rely on strong assumptions about the behavioral impacts of benefit sizes and the informal labor market. Thus, treating these results as a suggestion of the direction and relative magnitude of impact, it is clear that a transition from NAP to SNAP would present a decrease in overall employment while increasing benefit receipt. This provides evidence that employment behavior in Puerto Rico is generally in alignment with the known literature that higher welfare benefit amounts disincentivize labor.

Simulator Tool

A simulator tool is available for both replicating the results of this analysis and predicting the estimated impacts of a custom defined food stamps program in Puerto Rico. The simulator takes in the analysis year, income eligibility cutoff schedule, and maximum benefit schedule for a food stamp program and returns the estimated impacts on employment, weekly hours worked, and benefit take up. As an extension, this simulator can also be used to analyze counterfactual food stamp programs for any U.S. state. Detailed information and instructions for use are available in the simulator file of the Github for this project: <https://github.com/tgruslova/Food-Stamp-Programs-CSEC-4910>.

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Appendix

Income Variables for SimBenefit Calculation

Table 11: Employment Regression Coefficients for Men (Scaled by 100)

	HHIncome		Avg_HHIncome	
	<i>Employment</i>	<i>AME</i>	<i>Employment</i>	<i>AME</i>
<i>SimBenefit</i>	−0.0200 *** (0.0003)	−0.0021	−0.0200 *** (0.0004)	−0.0022
<i>EDU2</i>	11.380 *** (1.000)	1.4600	10.340 *** (0.900)	1.3600
<i>EDU3</i>	17.230 *** (1.000)	2.2100	21.600 *** (0.900)	2.8300
<i>EDU4</i>	20.760 *** (1.000)	2.6600	36.480 *** (1.000)	4.7800
<i>EDU5</i>	44.970 *** (1.400)	5.7600	45.610 *** (1.400)	5.9800
<i>EDU2</i> × <i>BLW2</i>	−13.470 *** (1.500)	−1.7300	−13.510 *** (1.500)	−1.7700
<i>SimBen</i> × <i>EDU2</i>	−0.0100 *** (0.0004)	−0.0017	−0.0072 *** (0.0005)	−0.0009
<i>SimBen</i> × <i>EDU3</i>	−0.0200 *** (0.0005)	−0.0023	−0.0200 *** (0.0005)	−0.0020
<i>SimBen</i> × <i>EDU4</i>	−0.0200 *** (0.0007)	−0.0024	−0.0100 *** (0.0005)	−0.0016
<i>SimBen</i> × <i>EDU5</i>	−0.0300 *** (0.0011)	−0.0041	−0.0200 *** (0.0013)	−0.0027
<i>Unemp</i>	−5.7200 *** (0.100)	−0.7300	−5.7900 *** (0.100)	−0.7600
<i>Age</i>	6.1800 *** (0.100)	0.7900	6.7000 *** (0.100)	0.8800
<i>Age</i> ²	−0.0600 *** (0.0010)	−0.0082	−0.0700 *** (0.0010)	−0.0088
<i>Married</i>	41.520 *** (0.600)	5.3200	36.790 *** (0.600)	4.8200
<i>CNST</i>	30.310 *** (2.200)		7.8500 *** (2.200)	
No. obs.	5048		5048	

Table 12: Employment Regression Coefficients for Women (Scaled by 100)

	HHIncome		Avg_HHIncome	
	<i>Employment</i>	<i>AME</i>	<i>Employment</i>	<i>AME</i>
<i>SimBenefit</i>	−0.0300 *** (0.0005)	−0.0035	−0.0300 *** (0.0007)	−0.0037
<i>EDU2</i>	39.360 *** (1.400)	4.6800	38.660 *** (1.400)	5.0500
<i>EDU3</i>	60.710 *** (1.400)	7.2200	58.090 *** (1.300)	7.5900
<i>EDU4</i>	80.540 *** (1.400)	9.5800	85.750 *** (1.400)	11.200
<i>EDU5</i>	83.280 *** (1.600)	9.9100	79.410 *** (1.500)	10.370
<i>EDU2 × BLW2</i>	−7.8800 *** (1.700)	−0.9400	−37.760 *** (1.600)	−4.9300
<i>SimBen × EDU2</i>	−0.0028 *** (0.0006)	−0.0003	0.0028 *** (0.0007)	0.0004
<i>SimBen × EDU3</i>	−0.0069 *** (0.0006)	−0.0008	0.0056 *** (0.0007)	0.0007
<i>SimBen × EDU4</i>	−0.0100 *** (0.0007)	−0.0012	−0.0020 *** (0.0008)	−0.0003
<i>SimBen × EDU5</i>	−0.0500 *** (0.0014)	−0.0055	0.0022 (0.0013)	0.0003
<i>Unemp</i>	3.3000 *** (0.200)	0.3900	2.980 *** (0.100)	0.3900
<i>Age</i>	4.2400 *** (0.100)	0.5000	3.240 *** (0.100)	0.4200
<i>Age²</i>	−0.0300 *** (0.0014)	−0.0039	−0.0200 *** (0.0013)	−0.0029
<i>Married</i>	12.330 *** (0.800)	1.4700	19.600 *** (0.700)	2.5600
<i>NOC</i>	15.190 *** (0.400)	1.8100	−0.8700 ** (0.400)	−0.1100
<i>OCUnder6</i>	18.550 *** (1.200)	2.2100	10.390 *** (1.100)	1.3600
No. obs.	4895		4895	

	HHIncome		Avg HHIncome	
	<i>Employment</i>	<i>AME</i>	<i>Employment</i>	<i>AME</i>
<i>FER</i>	27.280 *** (2.500)	3.2400	-7.7600 *** (2.000)	-1.0100
<i>NOC</i> $\times$ <i>Married</i>	10.030 *** (0.900)	1.1900	9.0700 *** (0.800)	1.1180
<i>OCUnder6</i> $\times$ <i>Married</i>	-37.180 *** (2.400)	-4.4200	-15.540 *** (2.200)	-2.0300
<i>FER</i> $\times$ <i>Married</i>	9.9200 * (5.100)	1.1800	42.000 *** (2.000)	5.4800
<i>CNST</i>	-39.880 *** (2.600)		-22.760 *** (2.500)	
No. obs.	4895		4895	

Table 13: Hours Worked Regression Coefficients for Men

	HHIncome	Avg_HHIncome
	WKHP	WKHP
<i>SimBenefit</i>	−0.0015 *** (0.0000)	−0.0016 *** (0.0000)
<i>WageIncome</i>	0.00005 *** (0.000004)	0.00005 *** (0.000004)
<i>EDU2</i>	1.7657 *** (0.5880)	1.4622 ** (0.5860)
<i>EDU3</i>	1.2071 ** (0.5960)	1.3909 ** (0.5910)
<i>EDU4</i>	2.2065 *** (0.6200)	2.2870 *** (0.6180)
<i>EDU5</i>	2.6798 *** (0.7400)	2.6342 *** (0.7410)
<i>EDU2 × BLW2</i>	−2.4762 ** (1.2290)	−2.6435 ** (1.2470)
<i>SimBen × EDU2</i>	−0.0008 ** (0.0000)	0.0011 ** (0.0000)
<i>SimBen × EDU3</i>	0.0001 (0.0000)	0.0011 ** (0.0010)
<i>SimBen × EDU4</i>	0.0003 (0.0010)	0.0027 *** (0.0010)
<i>SimBen × EDU5</i>	−0.0010 (0.0010)	0.0023 * (0.0010)
<i>Age</i>	0.7481 *** (0.0590)	0.7488 *** (0.0600)
<i>Age²</i>	−0.0084 *** (0.0010)	−0.0083 *** (0.0010)
<i>IND2</i>	1.2249 ** (0.5650)	1.3160 ** (0.5780)
<i>IND3</i>	0.2690 (0.4830)	0.2236 (0.4930)
<i>IND4</i>	0.6037 (0.6700)	0.7950 (0.6830)
No. obs.	4963	4963

	HHIncome	Avg_HHIncome
	<i>WKHP</i>	<i>WKHP</i>
<i>IND5</i>	−0.5059 (0.4400)	−0.5976 (0.4500)
<i>IND6</i>	−1.5729 *** (0.4850)	−1.4681 *** (0.4960)
<i>Married</i>	0.8572 *** (0.3180)	0.7544 ** (0.3230)
<i>CNST</i>	19.4568 *** (1.3780)	18.7576 *** (1.3970)
No. obs.	4963	4963

Table 14: Hours Worked Regression Coefficients for Women

	HHIncome	Avg_HHIncome
	<i>WKHP</i>	<i>WKHP</i>
<i>SimBenefit</i>	−0.0025 *** (0.0010)	0.0012 * (0.0010)
<i>WageIncome</i>	0.00007 *** (0.000005)	0.00008 *** (0.000005)
<i>EDU2</i>	−0.5970 (1.0260)	1.3886 (1.0330)
<i>EDU3</i>	1.2743 (0.9950)	3.5685 *** (1.0030)
<i>EDU4</i>	2.6707 *** (0.9980)	5.1185 *** (1.0050)
<i>EDU5</i>	2.8140 *** (1.0370)	5.0457 *** (1.0480)
<i>EDU2 × BLW2</i>	−2.2440 * (1.3230)	−4.1188 ** (1.3300)
<i>SimBen × EDU2</i>	0.0005 (0.0010)	−0.0021 *** (0.0010)
<i>SimBen × EDU3</i>	0.0002 (0.0010)	−0.0025 *** (0.0010)
<i>SimBen × EDU4</i>	0.0006 (0.0010)	−0.0022 *** (0.0010)
<i>SimBen × EDU5</i>	0.0006 (0.0010)	−0.0001 (0.0010)
<i>Age</i>	0.6444 *** (0.0670)	0.6280 *** (0.0680)
<i>Age</i> ²	−0.0072 *** (0.0010)	−0.0070 *** (0.0010)
<i>IND2</i>	1.5411 (1.1000)	1.6212 (1.1190)
<i>IND3</i>	−2.9024 *** (1.0380)	−2.5285 ** (1.0550)
<i>IND4</i>	0.1666 (1.0860)	0.4924 (1.1040)
No. obs.	4812	4812

	HHIncome	Avg_HHIncome
	<i>WKHP</i>	<i>WKHP</i>
<i>IND5</i>	−1.3991 (1.0160)	−1.0524 (1.0320)
<i>IND6</i>	−1.5450 (0.9980)	−1.3166 (1.0150)
<i>Married</i>	−0.1124 (0.3710)	−0.0888 (0.3770)
<i>NOC</i>	0.2385 (0.2560)	−0.3672 (0.2550)
<i>OCUnder6</i>	−0.4032 (0.5740)	−0.2569 (0.5830)
<i>FER</i>	0.9442 (1.2490)	0.3185 (1.2690)
<i>NOC × Married</i>	−0.2144 (0.3680)	0.1054 (0.3730)
<i>FER × Married</i>	−1.6819 (1.9170)	−0.9793 (1.9460)
<i>CNST</i>	20.6722 *** (1.8340)	18.1637 *** (1.8790)
No. obs.	4812	4812

Table 15: Take up Rate Regression Coefficients for Men (Scaled by 100)

	HHIncome		Avg_HHIncome	
	<i>Takeup</i>	<i>AME</i>	<i>Takeup</i>	<i>AME</i>
<i>SimBenefit</i>	0.0069 *** (0.0002)	0.0017	0.0035 *** (0.0002)	0.0008
<i>HHSize</i>	8.6200 *** (0.300)	2.0800	10.530 *** (0.200)	2.5500
<i>R65</i>	-15.640 *** (0.500)	-3.7700	-19.100 *** (0.600)	-4.6200
<i>EDU2</i>	-17.630 *** (1.200)	-4.2500	-15.720 *** (1.000)	-3.8100
<i>EDU3</i>	-44.640 *** (1.300)	-10.760	-50.910 *** (1.100)	-12.330
<i>EDU4</i>	-48.970 *** (2.100)	-11.800	-60.140 *** (1.700)	-14.560
<i>EDU5</i>	-143.94 *** (4.300)	-34.700	-122.27 *** (2.900)	-29.610
<i>SimBen</i> $\times$ <i>EDU2</i>	0.0017 *** (0.0004)	0.0004	0.0008 *** (0.0003)	0.0002
<i>SimBen</i> $\times$ <i>EDU3</i>	-0.0019 *** (0.0004)	-0.0005	0.0010 *** (0.0003)	0.0002
<i>SimBen</i> $\times$ <i>EDU4</i>	-0.0100 *** (0.0007)	-0.0026	-0.0062 *** (0.0005)	-0.0015
<i>SimBen</i> $\times$ <i>EDU5</i>	0.0100 *** (0.0015)	0.0036	0.0100 *** (0.0010)	0.0024
<i>Age</i>	0.3300 *** (0.100)	0.0800	0.2100 *** (0.100)	0.0500
<i>Age</i> ²	-0.0005 (0.0006)	-0.0001	-0.0002 (0.0006)	-0.00004
<i>Married</i>	-23.700 *** (0.700)	-5.7100	-23.750 *** (0.700)	-5.7500
<i>FS_Availability</i>	427.40 *** (10.100)	103.03	406.45 *** (10.100)	98.410
<i>FS_LagReceipt</i>	-0.0400 *** (0.0012)	-0.0087	-0.0400 *** (0.0012)	-0.0086
<i>CNST</i>	51.090 *** (1.700)	—	61.640 *** (1.700)	—
No. obs.	2910		2910	

Table 16: Take up Rate Regression Coefficients for Women (Scaled by 100)

	HHIncome		Avg_HHIncome	
	<i>Takeup</i>	<i>AME</i>	<i>Takeup</i>	<i>AME</i>
<i>SimBenefit</i>	0.0099 *** (0.0002)	0.0021	0.0049 *** (0.0002)	0.0011
<i>HHSIZE</i>	3.0100 *** (0.300)	0.6300	5.2600 *** (0.200)	1.1200
<i>R65</i>	-7.1200 *** (0.600)	-1.5000	-9.4800 *** (0.600)	-2.0100
<i>EDU2</i>	-15.750 *** (1.200)	-3.3200	-15.450 *** (1.100)	-3.2800
<i>EDU3</i>	-37.990 *** (1.200)	-8.0000	-24.980 *** (1.100)	-5.3100
<i>EDU4</i>	-96.670 *** (1.500)	-20.350	-90.600 *** (1.300)	-19.250
<i>EDU5</i>	-88.320 *** (2.600)	-18.590	-151.65 *** (2.100)	-32.220
<i>SimBen</i> × <i>EDU2</i>	-0.0056 *** (0.0003)	-0.0012	-0.0044 *** (0.0003)	-0.0009
<i>SimBen</i> × <i>EDU3</i>	-0.0019 *** (0.0004)	-0.0004	-0.0066 *** (0.0003)	-0.0014
<i>SimBen</i> × <i>EDU4</i>	-0.0012 ** (0.0005)	-0.0003	-0.0025 *** (0.0004)	-0.0005
<i>SimBen</i> × <i>EDU5</i>	-0.0300 *** (0.0009)	-0.0059	0.0033 *** (0.0007)	0.0007
<i>Age</i>	3.9900 *** (0.100)	0.8400	3.7700 *** (0.100)	0.8000
<i>Age</i> ²	-0.0400 *** (0.0006)	-0.0083	-0.0400 *** (0.0006)	-0.0080
<i>Married</i>	-42.970 *** (0.800)	-9.0400	-43.160 *** (0.800)	-9.1700
<i>FS_Availability</i>	482.92 *** (9.400)	101.64	449.92 *** (9.300)	95.590
<i>FS_LagReceipt</i>	-0.0400 *** (0.0011)	-0.0074	-0.0300 *** (0.0011)	-0.0073
No. obs.	3896		3896	

	HHIncome		Avg_HHIncome	
	<i>Takeup</i>	<i>AME</i>	<i>Takeup</i>	<i>AME</i>
<i>NOC</i>	7.0800 *** (0.400)	1.4900	7.9300 *** (0.400)	1.6900
<i>OCUnder6</i>	44.540 *** (1.100)	9.3700	46.530 *** (1.100)	9.8900
<i>FER</i>	−0.2900 (2.400)	−0.0600	−3.1600 (2.000)	−0.6700
<i>NOC</i> × <i>Married</i>	0.9000 (0.800)	0.1900	0.8900 (0.800)	0.1900
<i>OCUnder6</i> × <i>Married</i>	7.0600 ** (2.900)	1.4900	5.8400 ** (2.800)	1.2400
<i>FER</i> × <i>Married</i>	−58.700 *** (5.000)	−12.360	−74.970 *** (4.900)	−15.930
<i>CNST</i>	7.1700 *** (1.800)	—	20.490 *** (1.700)	—
No. obs.	3896		3896	

Results Using 2024 Data

Table 17: Employment Regression Coefficients for Men and Women (Scaled by 100)

	Men		Women	
	<i>Employment</i>	<i>AME</i>	<i>Employment</i>	<i>AME</i>
<i>SimBenefit</i>	−0.0100 *** (0.0004)	−0.0018	−0.0200 *** (0.0006)	−0.0029
<i>EDU2</i>	25.390 *** (0.900)	3.1800	51.780 *** (1.200)	6.0900
<i>EDU3</i>	36.310 *** (0.900)	4.5500	71.110 *** (1.100)	8.3700
<i>EDU4</i>	51.850 *** (1.000)	6.5000	101.39 *** (1.200)	11.930
<i>EDU5</i>	46.910 *** (1.300)	5.8800	105.47 *** (1.400)	12.410
<i>EDU2 × BLW2</i>	−31.140 *** (1.400)	−3.9000	−29.230 *** (1.700)	−3.4400
<i>SimBen × EDU2</i>	−0.0075 *** (0.0005)	−0.0009	0.0056 *** (0.0006)	0.0007
<i>SimBen × EDU3</i>	−0.0095 *** (0.0005)	−0.0012	−0.0091 *** (0.0006)	−0.0011
<i>SimBen × EDU4</i>	−0.0300 *** (0.0005)	−0.0035	0.0056 *** (0.0007)	0.0007
<i>SimBen × EDU5</i>	−0.0300 *** (0.0008)	−0.0036	−0.0074 *** (0.0009)	−0.0009
<i>Unemp</i>	−1.5200 *** (0.100)	−0.1900	−1.2500 *** (0.100)	−0.1500
<i>Age</i>	5.0900 *** (0.100)	0.6400	1.2000 *** (0.100)	0.1400
<i>Age²</i>	−0.0500 *** (0.0011)	−0.0058	0.0033 ** (0.0014)	0.0004
<i>Married</i>	38.800 *** (0.600)	4.8600	12.030 *** (0.800)	1.4200
<i>NOC</i>	—	—	7.0500 *** (0.400)	0.8300
No. obs.	4940		4674	

	Men		Women	
	<i>Employment</i>	<i>AME</i>	<i>Employment</i>	<i>AME</i>
<i>OCUnder6</i>	—	—	−28.450 *** (1.100)	−3.3350
<i>FER</i>	—	—	6.9800 *** (1.900)	0.8200
<i>NOC</i> × <i>Married</i>	—	—	11.150 *** (0.900)	1.3100
<i>OCUnder6</i> × <i>Married</i>	—	—	82.820 *** (3.600)	9.7400
<i>FER</i> × <i>Married</i>	—	—	−85.990 *** (4.000)	−10.120
<i>CNST</i>	−0.8600 (2.200)	—	28.780 *** (2.500)	—
No. obs.	4940		4674	

Table 18: Hours Worked Regression Coefficients for Men and Women

	Men	Women
	<i>WKHP</i>	<i>WKHP</i>
<i>SimBenefit</i>	−0.0016 *** (0.000)	−0.0008 (0.001)
<i>WageIncome</i>	0.00006 *** (0.000004)	0.00007 *** (0.000004)
<i>EDU2</i>	2.4196 *** (0.567)	0.8046 (0.864)
<i>EDU3</i>	2.7147 *** (0.570)	2.2132 *** (0.830)
<i>EDU4</i>	3.5252 *** (0.603)	4.0770 *** (0.834)
<i>EDU5</i>	2.6837 *** (0.728)	3.6113 *** (0.881)
<i>EDU2</i> × <i>BLW2</i>	0.8836 (1.188)	−1.4223 (1.361)
<i>SimBen</i> × <i>EDU2</i>	0.0010 ** (0.000)	−0.0006 (0.001)
<i>SimBen</i> × <i>EDU3</i>	0.0014 *** (0.000)	0.0009 (0.001)
<i>SimBen</i> × <i>EDU4</i>	0.0016 * (0.001)	0.00004 (0.001)
<i>SimBen</i> × <i>EDU5</i>	0.0018 (0.002)	−0.00009 (0.001)
<i>Age</i>	0.7751 *** (0.060)	0.8031 *** (0.066)
<i>Age</i> ²	−0.0087 *** (0.001)	−0.0088 *** (0.001)
<i>IND2</i>	2.3528 *** (0.579)	0.6065 (1.120)
<i>IND3</i>	0.0671 (0.469)	−3.5799 *** (1.056)
<i>IND4</i>	1.8026 *** (1.8026)	−0.8433 (1.115)
No. obs.	4869	4617

	Men	Women
	<i>WKHP</i>	<i>WKHP</i>
<i>IND5</i>	−0.0425 * (0.437)	−1.9938 * (1.033)
<i>IND6</i>	−0.8807 * (0.479)	−2.4163 ** (1.017)
<i>Married</i>	0.9074 *** (0.315)	−0.1435 (0.371)
<i>NOC</i>	—	−0.0291 (0.260)
<i>OCUnder6</i>	—	−1.2320 * (0.638)
<i>FER</i>	—	4.2448 *** (1.273)
<i>NOC × Married</i>	—	−0.2575 (0.391)
<i>FER × Married</i>	—	−1.3035 (2.181)
<i>CNST</i>	16.4766 *** (1.389)	16.1231 *** (1.823)
No. obs.	4869	4617

Table 19: Take up Rate Regression Coefficients for Men and Women (Scaled by 100)

	Men		Women	
	<i>Takeup</i>	<i>AME</i>	<i>Takeup</i>	<i>AME</i>
<i>SimBenefit</i>	0.0093 *** (0.0002)	0.0024	0.0036 *** (0.0002)	0.0008
<i>HHSize</i>	15.780 *** (0.300)	4.0300	14.220 *** (0.300)	3.3100
<i>R65</i>	-3.6400 *** (0.600)	-0.9300	-12.360 *** (0.600)	-2.8800
<i>EDU2</i>	-21.150 *** (1.000)	-5.4000	-37.720 *** (1.100)	-8.7700
<i>EDU3</i>	-40.970 *** (1.100)	-10.450	-47.870 *** (1.100)	-11.140
<i>EDU4</i>	-63.730 *** (1.700)	-16.260	-74.300 *** (1.400)	-17.280
<i>EDU5</i>	-87.820 *** (3.00)	-22.410	-166.39 *** (2.200)	-38.710
<i>SimBen</i> $\times$ <i>EDU2</i>	-0.0025 *** (0.0003)	-0.0006	0.0030 *** (0.0003)	0.0007
<i>SimBen</i> $\times$ <i>EDU3</i>	-0.0045 *** (0.0003)	-0.0012	-0.0017 *** (0.0003)	-0.0004
<i>SimBen</i> $\times$ <i>EDU4</i>	-0.0070 *** (0.0006)	-0.0018	-0.0030 *** (0.0004)	-0.0007
<i>SimBen</i> $\times$ <i>EDU5</i>	-0.0001 (0.0010)	-0.00004	0.0200 *** (0.0008)	0.0036
<i>Age</i>	2.0700 *** (0.100)	0.5300	3.3900 *** (0.100)	0.7900
<i>Age</i> ²	-0.0200 *** (0.0006)	-0.0054	-0.0300 *** (0.0006)	-0.0068
<i>Married</i>	-31.660 *** (0.800)	-8.0800	-52.410 *** (0.800)	-12.190
<i>FS_Availability</i>	428.61 *** (9.100)	109.37	417.45 *** (8.500)	97.110
<i>FS_LagReceipt</i>	-0.0300 *** (0.0014)	-0.0069	-0.0200 *** (0.0012)	-0.0051
No. obs.	2467		3285	

	Men		Women	
	<i>Takeup</i>	<i>AME</i>	<i>Takeup</i>	<i>AME</i>
<i>NOC</i>	—	—	0.8100 ** (0.400)	0.1900
<i>OCUnder6</i>	—	—	20.870 *** (1.000)	4.8600
<i>FER</i>	—	—	−8.6200 *** (2.300)	−2.0100
<i>NOC</i> × <i>Married</i>	—	—	−10.800 *** (0.800)	−2.5100
<i>OCUnder6</i> × <i>Married</i>	—	—	30.480 *** (2.800)	7.0900
<i>FER</i> × <i>Married</i>	—	—	−22.280 *** (5.600)	−5.1800
<i>CNST</i>	−15.470 *** (1.600)	—	−11.340 *** (1.600)	—
No. obs.	2467		3285	
